

The National Higher School of Artificial Intelligence
Introduction to AI
QUIZ #1 16/09/2023
Duration: 40 minutes

READ CAREFULLY:

- 1. Each of the 21 marks are marked out of 1 mark each; the last 4 questions are marked out of 1.5 marks each. Each incorrect answer is penalised 0.5 marks.**
- 2. For any question, there is ONLY ONE CORRECT ANSWER.**
- 3. N.B.: The questions may appear to you in a different order than your classmates.**

QUESTIONS:

1. Which search uses the problem specific knowledge beyond the definition of the problem?
a) Informed search
b) Depth-first search
c) Breadth-first search
d) Uninformed search
2. What is an AI 'agent'?
a) Takes input from the surroundings and uses its intelligence and performs the desired operations
b) An embedded program controlling line following robot
c) Perceives its environment through sensors and acting upon that environment through actuators
d) All of the above
3. What is meant by agent's percept sequence?
a) Used to perceive the environment
b) Complete history of actuator
c) Complete history of perceived things
d) None of the above
4. Which type of agent deals with happy and unhappy states?
a) Simple reflex agent
b) Model based agent
c) Learning agent
d) Utility based agent
5. Which of the following is not a type of Artificial Intelligence agent?
a) Learning AI agent
b) Goal-based AI agent
c) Simple reflex AI agent
d) Unity-based AI agent

6. Which family of search strategies is also called as Blind Search?
a) Uninformed search
b) Informed search
c) Simple reflex search
d) All of the above
7. What is the function of an Artificial Intelligence “Agent”?
a) Mapping of goal sequence to an action
b) Work without the direct interference of the people
c) Mapping of precept sequence to an action
d) Mapping of environment sequence to an action
8. Which search is implemented with an empty first-in-first-out queue?
a) Depth-first search
b) Breadth-first search
c) Bidirectional search
d) None of the above
9. Which of the following can improve the performance of an AI agent?
a) Perceiving
b) Learning
c) Observing
d) All of the above
10. What is the rule of simple reflex agent?
a) Simple-action rule
b) Condition-action rule
c) Simple & Condition-action rule
d) None of the above
11. When is breadth-first search optimal?
a) When there are less nodes from one level to the next one
b) When all step costs are equal
c) When all step costs are decreasing from one level to the next one.
d) None of the above
12. Given that b , l and m have the meanings seen in class, what is the worst-case space complexity of Depth-first search?
a) $O(b)$
b) $O(b \cdot l)$
c) $O(m)$
d) $O(b \cdot m)$

13. Which instruments are used for perceiving and acting upon the environment?

- a) Sensors and Actuators
- b) Sensors
- c) Perceiver
- d) None of the above

14. Which element in the agent are used for selecting external actions?

- a) Perceive
- b) Performance
- c) Learning
- d) Actuator

15. Which search algorithm imposes a fixed depth limit on nodes?

- a) Depth-first search
- b) Depth-limited search
- c) Iterative deepening search
- d) Bidirectional search

16. Which search uses a stack for searching the states?

- a) A* search
- b) Depth-first search
- c) Breadth-first search
- d) None of the above

17. What is the other name for Heuristic Search Strategy?

- a) Uninformed search
- b) Informed search
- c) Blind search
- d) None of the above

18. What is the heuristic function of Greedy Best-First Search?

- a) $f(n) = g(n) + h(n)$
- b) $f(n) < h(n)$
- c) $f(n) = h(n)$
- d) $f(n) < g(n) + h(n)$

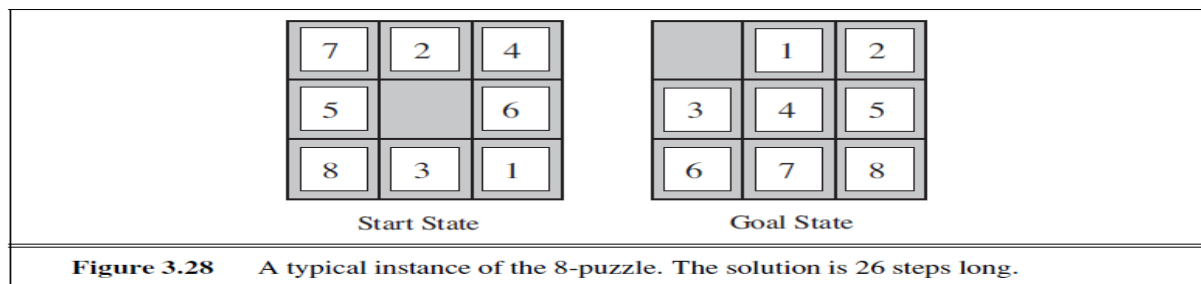
19. Which of the following search algorithms has linear space complexity?

- a) Best-first search
- b) Recursive best-first search
- c) Breadth-first search
- d) None of the above

20. Which method is used to search better by learning?
- a) Best-first search
 - b) A* search
 - c) Meta-level state space
 - d) None of the above
21. Which search method will expand the node that is closest to the goal?
- a) Best-first search
 - b) Greedy best-first search
 - c) A* search
 - d) None of the above

Questions 22 to 25 use the following 8-puzzle problem:

Consider the 8-puzzle problem shown in the following figure where the initial state is on the left, the goal state on the right and the possible actions are the blank tile to go right, down, left, or up (in this order if any search is attempted). Considering that the 8-puzzle is a 3x3 matrix, where (i,j) means tile at row i and column j.



22. Applying Depth-First Search, what is the position of the grey (empty) tile in the last node generated after a total of 4 nodes generated?
- a) (1,3)
 - b) (3,3)
 - c) (2,3)
 - d) (3,1)
 - e) None of the above
23. Applying Breadth-First Search, what is the position of the grey (empty) tile in the last node generated after a total of 7 nodes generated?
- a) (1,3)
 - b) (3,3)
 - c) (1,1)
 - d) (3,1)
 - e) None of the above

24. Applying Depth-Limited Search with $L=2$, what is the position of the grey (empty) tile in the last node generated?
- a) (1,3)
 - b) (3,3)
 - c) (1,1)
 - d) (3,1)
 - e) None of the above
25. Suppose we use a heuristic function $h =$ “number of misplaced tiles with respect to their expected positions in the goal”. Applying *Greedy Best First Search* using this heuristic, what is the position of the grey (empty) tile in the last node generated after a total of 6 nodes generated? In case of equality between nodes, select the leftmost node to expand.
- f) (1,2)
 - g) (2,3)
 - h) (1,1)
 - i) (2,2)
 - j) None of the above